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CFD-modeling of steam methane reforming via particle-resolved and pseudo-homogeneous models: comparative analysis with experimental verification

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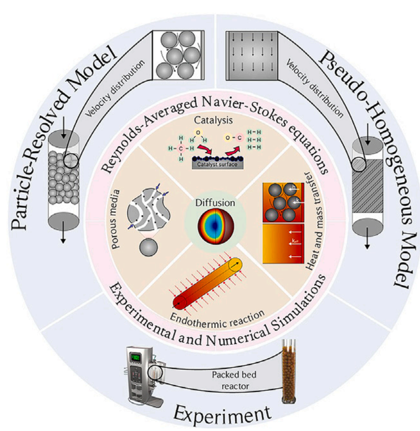
HIGHLIGHTS

- CFD-modeling of steam methane reforming via pseudo-homogeneous and particle-resolved models.
- Experimental results to verify the numerical results.
- Good agreement between the results for the case with adiabatic wall.
- Near-equilibrium methane conversion for the case with heat supply at T > 1200 K.
- Significant deviation in temperature contours for the case with heat supply for PRM and PHM.

GRAPHICAL ABSTRACT



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ABSTRACT

Steam methane reforming (SMR) is a widely used process for hydrogen and synthesis gas production. The experimental and numerical approaches are employed to understand and predict SMR performances. Among the numerical approaches, there are two main models: pseudo-homogeneous model (PHM) and particle-resolved model (PRM). In this paper, a comparative analysis of PHM and PRM for the numerical simulation of SMR was conducted. To verify the numerical results, experimental tests were performed in a lab-scale electrified reformer. Both numerical and experimental tests were carried out for two cases: with and without heat supply through the wall. It was established that PHM and PRM accurately predict temperature and methane conversion for the case with an adiabatic wall (no heat flux through the wall). The maximum deviation for temperature and methane conversion is less than 10 %. However, for the case with heat supply through the wall, the discrepancy between results obtained via PHM and PRM increases with rising temperature and residence time. For an inlet temperature of 1200K, the discrepancy can be as much as 200 K. Such discrepancies were explained, with one possible reason being the difference in the nature of heat transfer in cases with and without heat supply through the wall.

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1. Introduction

Steam methane reforming (SMR) is a widely used process for producing both hydrogen and synthesis gas [1]. SMR using renewable electricity is a promising direction for low-carbon hydrogen production [2,3]. Despite being well known since the 20th century, nowadays, the number of publications on this process is constantly increasing. The main reason for this is that the performance of SMR is affected by many parameters, both operational (pressure, temperature, inlet reactant composition, residence time, etc.) [4,5] and design-related (type and shape of catalyst, reformer, etc.) [6,7]. Moreover, steam methane reforming is a complex process that includes various physicochemical phenomena, such as heat and mass transfer, flow dynamics, chemical reactions, and diffusion [8].

The industrial steam methane reforming process occurs over various catalysts, among which Ni-based catalysts are widely used [9,10]. The Ni-based catalysts can vary in size, shape, composition, and types of supports. Typically, industrial Ni-based catalysts are porous particles with a size of 5–20 mm [11–13]. For such particles, SMR reactions occur inside the porous catalyst, and intra-particle diffusion strongly affects reaction performance [14]. For small-scale SMR, the catalyst can be crushed into small particles with a size of 0.1–0.2 mm; in this case, the reformer will operate under close to intrinsic kinetic conditions [15,16].

To understand the effect of operational and design-related parameters on SMR performance, both experimental and numerical studies are widely used. Experiments allow for establishing the dependence of temperature distribution, methane conversion, and hydrogen yield on temperature, pressure, and residence time for real industrial reformers [17–20]. However, there are often many safety restrictions regarding testing a wide range of operational and design parameters for reforming. Additionally, experiments on SMR require significant resources.

Numerical simulation of SMR has received wide interest among researchers in recent decades, mainly due to increased computational power and the development of modern tools for numerical simulations [21–24]. Numerical simulations of SMR are based on Computational Fluid Dynamics (CFD), which allows for obtaining comprehensive information about the process in various parts of reformers under different operational parameters for various designs of reformers. There are two main approaches for CFD modeling of SMR: the particle-resolved model (PRM) [25,26] and the pseudo-homogeneous model (PHM) [27,28].

The Particle-Resolved Model (PRM) considers each individual catalyst particle within a fixed-bed reactor. The research group led by Dixon at Worcester Polytechnic Institute has recently published several papers on the application of PRM for the numerical simulation of various chemical processes [29–31]. Notably, Dixon presented findings from a study on heat and mass transfer for steam methane reforming in a fixed-bed tubular reformer filled with 807 spherical particles [29]. The author derived temperature gradients, mole fractions, and reaction rates using Langmuir–Hinshelwood–Hougen–Watson kinetics from Hou and Hughes' paper [32]. Additionally, Dixon reported that the computational time for these simulations was approximately 30 days, utilizing a PowerEdge R610 server with Ansys Fluent [29].

In parallel, Lu and Nikrityuk developed and validated a new model for steam methane reforming based on the Discrete Element Method (DEM), which considers each catalyst particle in a fixed bed [33]. This model incorporates intra-particle heat and mass transfer for individual particles, coupled with bulk flow dynamics. Validation against Dixon's results demonstrated that this approach provides the accurate results with significantly reduced computational time to approximately 0.5 hours on an i7-5820HK 3.6 GHz computer using MATLAB.

One of the main limitations of PRM is that an increase in the number of particles leads to a corresponding increase in required computational time. Furthermore, the incorporation of various models into the CFD code also increases computational resources [26]. Consequently, applying the PRM approach to simulate an industrial-scale steam methane reformer with tens of thousands of catalyst particles presents significant challenges. To address this, the pseudo-homogeneous model (PHM) is commonly employed for simulating industrial-scale steam methane reforming, treating the fixed-bed reactor as a porous medium [34].

Tran et al. reported on CFD modeling of an industrial-scale steam methane reforming furnace comprising 336 reforming tubes, 96 burners, and 8 flue gas tunnels [19]. In their study, the catalytic packed bed was modeled as a porous zone with parameters estimated from the semi-empirical Ergun equation. Implemented in Ansys Fluent, their CFD model demonstrated that PHM can effectively simulate industrial-scale reformers while yielding accurate predictions for temperature, concentration, and pressure drop.

Tacchino et al. conducted a numerical study of a fixed-bed steam methane reforming reactor, validating the obtained results against industrial experimental data [20]. They developed a one-dimensional model that encompasses heat and mass transfer, flow dynamics, and diffusion. The Ergun correlation was utilized to establish the relationship between pressure drop and velocity in fluid flow through porous media. Their CFD model was executed using COMSOL, achieving an acceptable computational burden and time, with numerical results aligning well with experimental data.

The PHM approach offers a significant simplification of CFD models compared to PRM and requires less computation time. However, there exists a set of publications analyzing discrepancies between results obtained via PRM and PHM for various processes [35–37].

Both PRM and PHM can be used for CFD modeling of steam methane reforming with certain assumptions. One of the main assumptions in PRM involves defining contact points between catalyst particles [38]; typically, gaps between particles and walls are established to minimize errors in mesh generation and ensure stability in solutions. In contrast, the PHM approach models heat transfer from the hot wall to the packed bed through a porous zone. In this context, heat is conducted from the wall to the reaction space via thermal conductivity; however, in a real packed bed, heat transfer additionally occurs through convection [39].

The primary objective of this paper is to conduct a comparative analysis of particle-resolved (PRM) and pseudo-homogeneous (PHM) models for the numerical prediction of steam methane reforming performance. To facilitate this comparison and provide verification of the numerical results, additional experimental tests were conducted on a lab-scale fixed-bed reactor filled with industrial Ni-Al_2O_3 catalyst particles. The presented work is the first study where a comparative analysis of two models for predicting SMR process characteristics is carried out. In this work, for the first time, a series of experiments and numerical calculations for PHM and PRM for an isothermal reformer wall are presented, which allows for a clear understanding of the influence of the model on the results for different reformer operating modes.

2. Methodology

In this section, the methodology for the numerical simulation of steam methane reforming using the PRM and PHM approaches is described, including the computational domain and mesh, governing equations, and boundary conditions. Moreover, a procedure for the experimental tests is provided, including catalyst characteristics and equipment descriptions.

2.1. Numerical simulation

2.1.1. Computational domain and mesh

The numerical simulation was performed for both PHM and PRM cases. In PHM, the packed bed was set as a porous homogeneous medium, while in PRM, each catalyst particle was treated as a single particle. The parameters of the computational domain for both models are schematically presented in Fig. 1. The dimensions of the packed bed are the same for both computational domains: the length of the packed bed is 360 mm, the diameter is 60 mm, and the lengths of the inlet and outlet zones are 100 mm and 150 mm, respectively. The packed bed

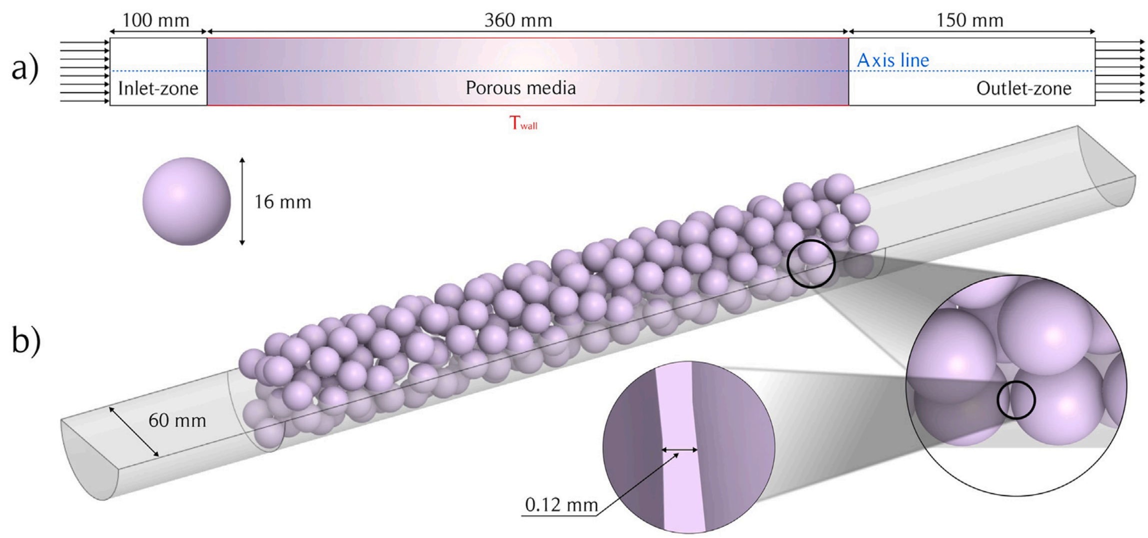


Fig. 1. The computational domain for a pseudo-homogeneous model (a) and a particle-resolved model (b).

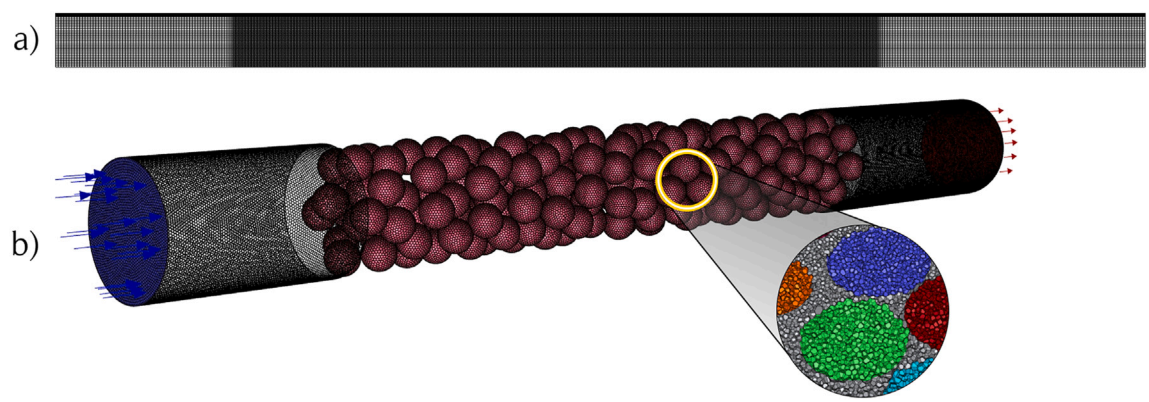


Fig. 2. The computational mesh for a pseudo-homogeneous model (a) and a particle-resolved model (b).

for PRM was generated for the reformer filled with spheres of 16.0 mm in diameter. To generate the packed bed, we used our own software **SAND** for the generation of randomly packed beds [40]. The particles of the packed bed are represented using a minimal number of facets. The particles were scaled down to 0.99 to avoid errors in mesh generation. Dixon et al. showed that for low Re (as used in our paper) such approach provides similar results to bridge and scale-up approaches [38].

The computational meshes for both models are presented in Fig. 2. The computational mesh for PHM was created for a 2 D-axisymmetric computational domain. The mesh for PHM is structured with inflation in the near-wall region to achieve a y^+ parameter below 5. The mesh for PRM is a hexahedral structured mesh. Compared with unstructured meshes, the use of structured meshes can improve calculation accuracy, save computational power, and decrease discrepancies in results. The mesh was generated for both the flow region (the zone between catalyst particles) and for the catalyst particles (the intra-particle zone) to consider intra-particle processes.

The number of elements for each case of computational mesh was determined based on a mesh independence study. A summary of this study is presented in Table 1. For the mesh independence study, five meshes were tested for each computational domain: the packed bed filled with spheres (PRM) and the packed bed set via the pseudo-homogeneous model (PHM). Methane conversion was used as the control parameter. It was established that an appropriate solution for the particle-resolved model can be achieved with approximately 10^7 mesh elements for the spherical particles, while for the pseudo-homogeneous model, the appropriate number of mesh elements is one order lower. The main reason for this is the simpler computational domain compared to that of the packed bed in the particle-resolved case.

2.1.2. Governing equations

The equations of mass, momentum, and energy conservation, coupled with chemical kinetics, diffusion, and porous medium models, were solved via ANSYS Fluent for steady-state conditions.

The main chemical reactions for steam methane reforming are provided below. A set of these chemical reactions is widely used in the kinetic models for SMR [16,32,41]:

$$CH_4 + H_2O = CO + 3H_2 - 206.1\,\text{kJ/mol}. \tag{1}$$

$$CO + H_2O = CO_2 + H_2 + 41.1\,\text{kJ/mol}. \tag{2}$$

$$CH_4 + 2H_2O = CO_2 + 4H_2 - 165.0\,\text{kJ/mol}. \tag{3}$$

Mass, momentum, energy conservation

The mass and momentum equations have the following form [42,43]:

$$\nabla \cdot (\rho \vec{v}) = 0; \quad \nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla \cdot p + \nabla \cdot (\bar{\bar{\tau}}) + \rho \vec{g} + \vec{F} + S_i \tag{4}$$

where p – pressure; $\rho \vec{g}$ – gravitational body force; $\vec{F}$ – external body forces; S_i – momentum source term for porous medium.

Table 1
The mesh independence study for the packed bed filled with spheres (PRM) and the packed bed set via the pseudo-homogeneous model (PHM): Elem – number of mesh elements; X_{CH_4} – methane conversion.

Case	1		2		3		4		5	
Models	Elem	X_{CH_4}	Elem	X_{CH_4}	Elem	X_{CH_4}	Elem	X_{CH_4}	Elem	X_{CH_4}
PRM	1.2×10^5	0.225	5.7×10^5	0.191	1.5×10^6	0.152	8.7×10^6	0.145	1.2×10^7	0.144
PHM	8.1×10^4	0.222	1.2×10^5	0.186	2.5×10^5	0.162	5.1×10^5	0.160	8.8×10^5	0.160

Stress tensor $\bar{\bar{\tau}}$ in Eq. (4) can be expressed as:

$$\bar{\bar{\tau}} = \mu\left[(\nabla\vec{v} + \vec{v}^{\mathrm{T}}) - \frac{2}{3}\nabla\cdot\vec{v}I\right] \tag{5}$$

where μ – molecular viscosity; I – unit tensor, and the second term on the right hand side is the effect of volume dilation.

The energy equation is based on the law of energy conservation and has the following form:

$$\nabla\cdot\vec{v}(p+\rho\cdot h_f) = \nabla\cdot\left[k_{eff}\cdot\nabla T - \left(\sum h_j\cdot\vec{J}_j\right) + \bar{\bar{\tau}}\cdot\vec{v}\right] + S_f^h \tag{6}$$

where T – reaction mixture temperature; h_f – reaction mixture enthalpy; $\vec{J}_j$ – diffusion flux of j element; k_{eff} – thermal conductivity; h_j – enthalpy of j component; S_f^h – the source of energy due to chemical reactions.

The effective thermal conductivity of the porous medium, k_{eff}, is defined as follows:

$$k_{\mathrm{eff}} = \epsilon_p k_f + (1-\epsilon_p)k_s \tag{7}$$

where k_f, k_s – fluid phase and solid medium thermal conductivity, respectively; ϵ_p – porosity of catalytic particle.

The P1 radiation model is used to account for the radiative heat flux [44].

The transition shear stress transport (SST) turbulence model is employed to solve the Reynolds-averaged Navier–Stokes (RANS) equations. [45].

Porous medium

The porous medium of the catalyst particles in the packed beds for the particle-resolved model is defined by the addition of a momentum source term (S_i) to Eq. (4). The source term consists of two parts: a viscous loss term and an inertial loss term. The actual pore size of the catalyst particles ranges from 10 to 100 nm. For such small pore sizes, the momentum transported into the particle is negligible. Therefore, the transport phenomena are defined by a permeability (α):

$$S_i = -\frac{\mu}{\alpha}v_i \tag{8}$$

The permeability (α) depends on pore size, tortuosity, etc. The permeability of the Ni-Al_2O_3 catalyst is set at $2.8\cdot10^{-14}$ m^2 [46].

The porous medium of the packed bed for the pseudo-homogeneous model is defined via simple homogeneous porous media expressed as:

$$S_i = -\left(\frac{\mu}{\alpha}v_i + C_2\frac{1}{2}\rho|v|v_i\right) \tag{9}$$

The source term in Eq. (9) consists of two parts: α is the permeability and C_2 is the inertial resistance factor. These terms were determined using the modified Ergun equation [47]. In our previous work [47], we derived the modified Ergun equation based on experiments to account for catalyst roughness at low tube-to-particle diameter ratios

Diffusion

In this paper, the diffusion is simulated via Bosanquet approximation [48] The effective diffusion of i-th species is based on Bosanquet approximation:

$$D_i^{eff} = \left(\frac{1}{D_{m,i}^{eff}} + \frac{1}{D_{i,k}^{eff}}\right)^{-1} \tag{10}$$

where $D_{m,i}^{eff}$ – effective ordinary diffusivity; $D_{i,k}^{eff}$ – effective Knudsen diffusivity.

Effective ordinary diffusivity ($D_{m,i}^{eff}$) is defined from:

$$D_{m,i}^{eff} = \frac{\epsilon}{\tau}\frac{1-x_i}{\sum\frac{x_i}{D_{ij}}} \tag{11}$$

where ϵ – porosity; τ – tortuosity; x_i – molar fraction of ith species; D_{ij} – binary diffusivity for i–jth pair.

The binary diffusivity (D_{ij}) is calculated from:

$$D_{ij} = \frac{0.0143T^{1.75}}{p_t M^{0.5}\left(V_i^{1/3} + V_j^{1/3}\right)^2} \tag{12}$$

where $V_{i,j}$ – diffusion volume for i,jth species (from [49]: H_2 = 7.07; H_2O = 12.7; CO_2 = 26; CO = 18.9; CH_4 = 16.8)

The effective Knudsen diffusivity is determined from:

$$D_{i,k}^{eff} = \frac{\epsilon}{\tau}48.5d_p\left(\frac{T}{M_i}\right)^{0.5} \tag{13}$$

where d_p – mean pore diameter.

For the pseudo-homogeneous model, to account for intra-particle diffusion limitations, the effectiveness factor (η) was used. In the author's previous publication [14], the equation for calculating the effectiveness factor as a function of catalyst particle size, temperature, and steam-to-methane ratio was obtained in the following form:

$$\eta = f_d(d)\cdot f_T\left(\frac{T}{1000}\right)\cdot f_b\left(\frac{N_{H_2O}}{N_{CH_4}}\right), \tag{14}$$

where $f_d(d)$ – the function of the effectiveness factor on diameter; $f_T\left(\frac{T}{1000}\right)$ – the function of the effectiveness factor on temperature; $f_b\left(\frac{N_{H_2O}}{N_{CH_4}}\right)$ – the function of the effectiveness factor on steam-to-methane ratio.

In Eq. (14), the functions $f_d(d)$, $f_T\left(\frac{T}{1000}\right)$, $f_b\left(\frac{N_{H_2O}}{N_{CH_4}}\right)$ are taken from the paper [14].

Kinetic model

The reactions (1)–(3) occur in porous catalysts, and their reaction rates are determined based on Hou's kinetic model, which is widely used for numerical simulations of steam methane reforming [32]:

$$R_1 = \frac{k_1(p_{\mathrm{CH_4}}p_{\mathrm{H_2O}}^{0.5} - p_{\mathrm{H_2}}^3 p_{\mathrm{CO}}/K_1 p_{\mathrm{H_2O}}^{0.5})}{p_{\mathrm{H_2}}^{1.25}} \times \frac{1}{\mathrm{DEN}^2} \tag{15}$$

$$R_2 = \frac{k_2(p_{\mathrm{CO}}p_{\mathrm{H_2O}}^{0.5} - p_{\mathrm{H_2}}p_{\mathrm{CO_2}}/K_2 p_{\mathrm{H_2O}}^{0.5})}{p_{\mathrm{H2}}^{0.5}} \times \frac{1}{\mathrm{DEN}^2} \tag{16}$$

Table 2
The coefficients for the equations of kinetic model.

Kinetics		
k_1	=	$5.922 \times 10^{8} \exp\left(\frac{-209200}{R_g T}\right)$
k_2	=	$6.028 \times 10^{-4} \exp\left(\frac{-15400}{R_g T}\right)$
k_3	=	$1.093 \times 10^{3} \exp\left(\frac{-109400}{R_g T}\right)$
Adsorption		
K_{H_2O}	=	$9.251 \times \exp\left(\frac{-15900}{R_g T}\right)$
K_{H_2}	=	$5.68 \times 10^{-10} \exp\left(\frac{93400}{R_g T}\right)$
K_{CO}	=	$5.127 \times 10^{-13} \exp\left(\frac{140000}{R_g T}\right)$
Equilibrium		
K_1	=	$1.198 \times 10^{17} \exp\left(\frac{-26830}{T}\right)$
K_2	=	$1.767 \times 10^{-2} \exp\left(\frac{4400}{T}\right)$
K_3	=	$2.117 \times 10^{15} \exp\left(\frac{-22430}{T}\right)$

$$R_3 = \frac{k_3(p_{CH_4} p_{H_2O} - {p_{H_2}}^4 p_{CO_2}/K_3 p_{H_2O})}{p_{H_2}^{3.5}} \times \frac{1}{\mathrm{DEN}^2} \tag{17}$$

$$\mathrm{DEN} = 1 + K_{CO} \cdot p_{CO} + K_H \cdot p_{H_2}^{0.5} + K_{H_2O} \cdot p_{H_2O}/p_{H_2} \tag{18}$$

The expressions for coefficients in (15)–(17) are presented in Table 2:

To implement both chemical kinetics and diffusion model in the Fluent solver, each model was converted into a User-Defined Function (UDF) written in C++ and uploaded to the Fluent solver.

2.1.3. Boundary conditions and solution procedure

The velocity inlet boundary conditions are set as follows: $u_x = u$, $v = 0$, $p_g = 0$. At the outlet, pressure outlet boundary conditions are applied. The reformer wall temperature for the packed bed zone was kept constant at T = 800, 1000, or1200 K. The walls at the inlet and outlet zones are set as adiabatic. The operating pressure is 120 kPa. Residence time was varied from 50 to 300 kgcat·s/kmol$_{CH_4}$. The inlet steam-to-methane ratio (mol/mol) is set at 2.0. The parameters of the packed bed and the properties of the catalyst particles are consistent with those used in the experimental part and are provided below.

The parameters of the Fluent solvers are as follows:

- Solver Type: Pressure-Based;
- Time: Steady State;
- Solution Scheme: SIMPLE;
- Spatial Discretization:
  - Gradient-Gauss Node Based;
  - Pressure: standard;
  - Momentum, Energy, Turbulence, Species: Third-Order MUSCL;
- Controls for Species and Energy: 0.95;
- Residual:
  - Energy: 1 e − 06;
  - Others: 1 e − 05;
- Initialization: Standard;

2.2. Experiment

Setup

The experimental setup for the study of steam methane reforming is classical. Fig. 3 shows a schematic representation and a photo of the experimental setup of universal electrified fixed bed furnace (Anhui Kemi Instrument Co., Ltd.). The main element of the setup is an electric furnace that can maintain a constant temperature of the reactor wall. The furnace consists of three equal heating zones, each 360 mm in length. The bottom heating zone is used to supply heat to the packed bed filled with catalyst pellets for steam methane reforming, while the upper and middle heating zones are used to preheat the steam–methane mixture. Steam for the reaction is generated using an electric steam generator and mixed with methane in a mixer. The reformer tube is made of quartz.

The reaction products after passing through the packed bed are divided into two streams. The first stream is fed to the exhaust gas system

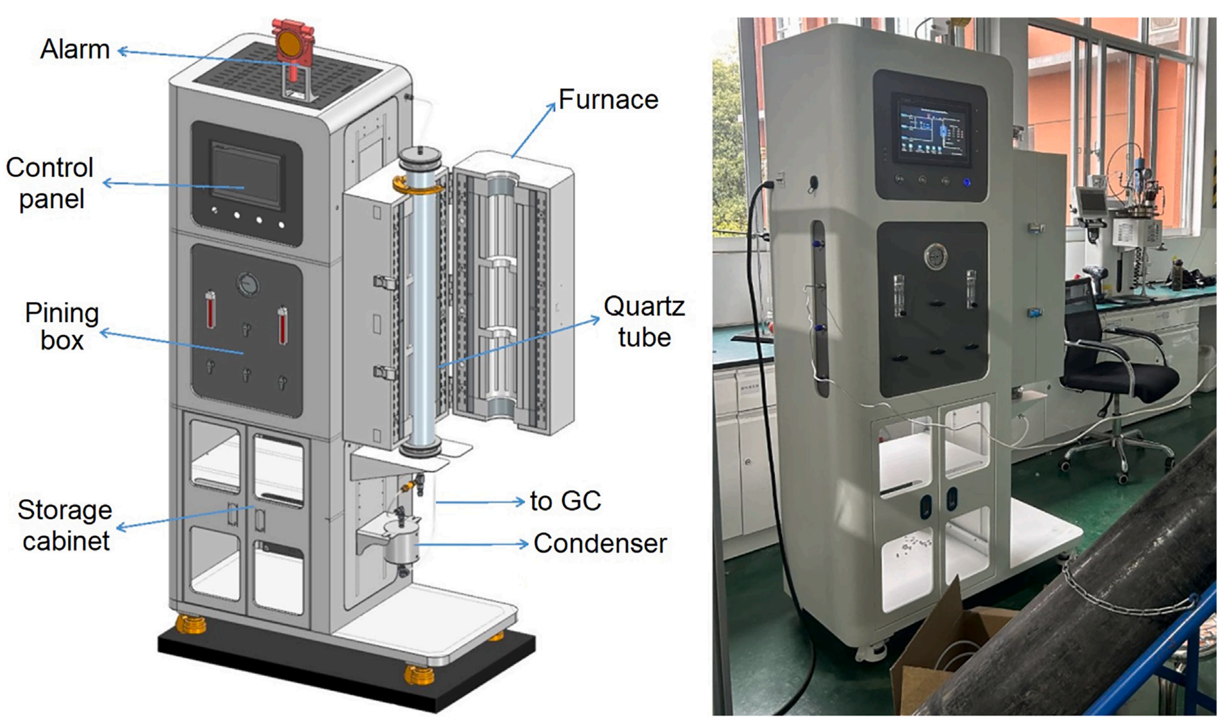


Fig. 3. The electrified furnace for experimental study produced by Anhui Kemi Instrument Co., Ltd.

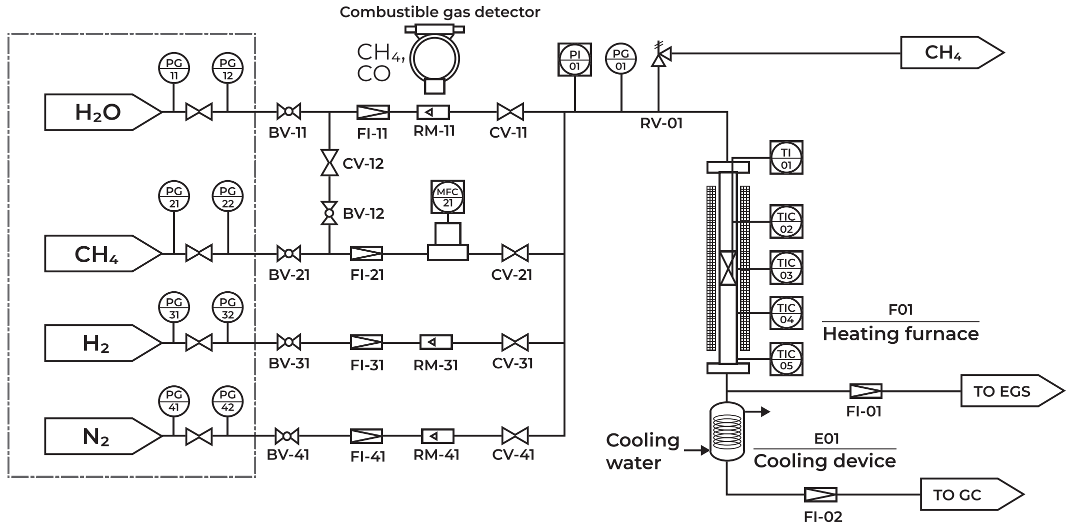


Fig. 4. The schematic diagram of the experimental setup: BV – Ball Valve, PG – Pressure Gauge, RM – Rotameter, GC – Gas Chromatography, EGS – Exhaust Gas System, FI – Filter, CV – Check Valve, MFC – Mass Flow Controller & Meter, TI – Temperature Transducer, TIC – Temperature Controller, PI – Pressure Transducer, RV – Safety Valve, SV – Electromagnetic Valve.

(EGS). The second stream is supplied to a condenser and then to gas chromatography (GC). The schematic diagram of the experimental setup is shown in Fig. 4.

Catalyst

An industrial Ni-based catalyst is used for the experiments. This catalyst consists of nickel oxide on a calcium aluminate support, typically containing 16 % NiO, 1.5 % SiO_3 and 0.05 % SO_3. The catalyst properties are as follows: surface area of 14.3 m^2/g; skeletal density of 3190 kg/m^3; porosity of 0.44. The catalysts are the same as those used in the experimental paper by Hou and Hughes [32]. The diameter of the spherical particles is shown in Fig. 1.

The catalyst pre-treatment followed a traditional procedure: the catalyst was heated to 500 °C at a rate of 5 °C/min in nitrogen and maintained at this temperature for 30 minutes. Then, the packed bed was filled with hydrogen at the same temperature and maintained for another 30 minutes. Afterward, the catalyst particles were heated to 700 °C at 5 °C/min and maintained at this temperature for 30 minutes. Following this, the temperature of the electric furnace was set to the required operating temperature. After pre-treatment, the packed bed was ready for experimental tests.

Experimental procedure

The experimental tests were performed for various cases. The first case involved steam methane reforming without heat supply via the reformer wall. The inlet mixture temperatures were set at 800, 1000, and 1200 K; pressure was maintained at 120 kPa; the steam-to-methane ratio was set at 2.0; and residence time varied in the range of approximately 50–300 $kg_{cat}{\cdot}s/kmol_{CH_4}$. The second case involved steam methane reforming with heat supply through the wall, where the wall temperature was equal to the inlet mixture temperature (800, 1000, or 1200 K). For this second case, other operational parameters remained the same as in the first case: pressure was 120 kPa; steam-to-methane ratio was 2.0; and residence time ranged from approximately 50–300 $kg_{cat}{\cdot}s/kmol_{CH_4}$. The residence time was adjusted by changing the flow rate of the inlet reaction mixture. It should be noted that the reaction mixture did not contain hydrogen; however, hydrogen from the pre-treatment procedure was used to initiate the chemical reactions. The running time of each test approximately 30 minutes after reaching a stable state.

The furnace consists of three equal sections. The first two upper sections were used to heat the steam–methane mixture up to the temperature of the reaction mixture at the inlet of the reaction zone. To achieve this, the upper sections were filled with inert ceramic foam.

During the experimental tests, temperature and methane conversion at the reformer outlet were measured for all cases. Other initial parameters were controlled via a control panel. The methane conversion (X_{CH_4}) is determined as follows:

$$X_{CH_4}\,(\%) = \frac{[CH_4]_{in} - [CH_4]_{out}}{[CH_4]_{in}} \times 100\ \% \tag{19}$$

where $[CH_4]_{in}$, $[CH_4]_{out}$ – mole of methane at the reformer inlet and outlet, respectively.

3. Results

3.1. Methane conversion

Numerical calculations and experimental tests were conducted for two cases: no heat supply through the wall of the reformer and heat supply through the wall to maintain a constant temperature of the reformer wall.

No heat supply through the wall

The methane conversion (X_{CH_4}) in the steam methane reforming process was experimentally measured and numerically calculated. The contours of the methane mass fraction obtained via the PHM (a) and PRM (b) without heat supply through the wall of the reformer filled with spherical particles at T = 1000 K, β = 2.0, and a residence time of 200 $kg_{cat}{\cdot}s/kmol_{CH_4}$ are shown in Fig. 5. Fig. 5 shows a non-monotonic decrease in CH_4 conversion along the axial direction for both pseudo-homogeneous and particle-resolved models, as depicted in the bottom part of Fig. 5. The main reason for this observation is the effect of temperature on reaction rates according to Eqs. (15)–(17). Thus, a decrease in temperature leads to a non-linear decrease in reaction rates. Fig. 5 (bottom) depicts the axial distribution of methane conversion for both PHM and PRM cases. For the PRM case, the finite volumes of the extra-particle flow region were considered to calculate the average CH_4 mass fraction. As seen in the bottom part of Fig. 5, PRM provides lower CH_4 conversion compared to PHM. This can be explained by the fact that in the particle-resolved case, the catalyst particles are not uniformly placed in the packed bed, leading to large gaps that result in lower contact time between the reaction mixture and the catalyst.

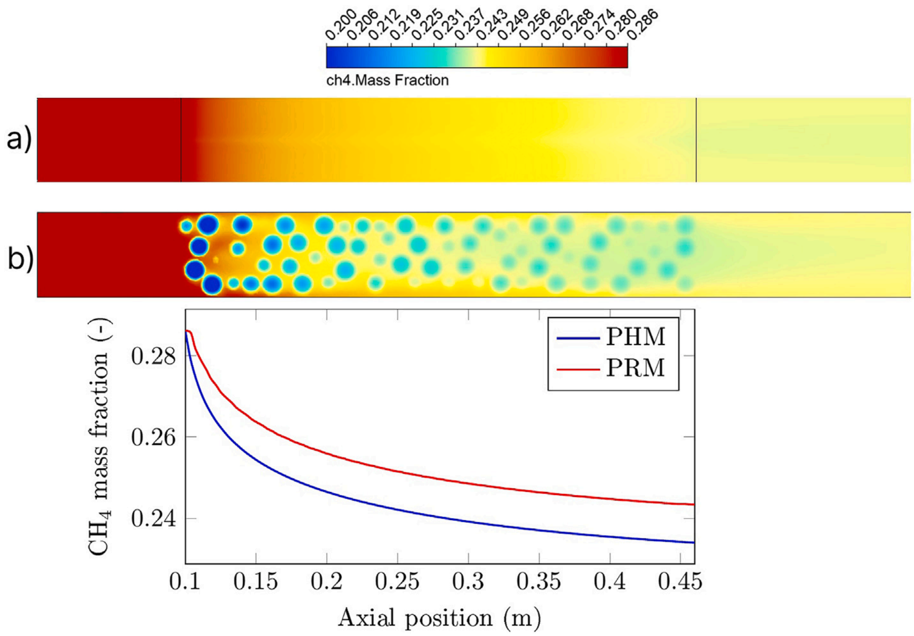


Fig. 5. The contours of methane mass fraction for PHM (a) and PRM (b) without heat supply through the wall of reformer filled with spherical particles at: T = 1000 K, β = 2.0, residence time of 200 $kg_{cat}{\cdot}s/kmol_{CH_4}$. Bottom: axial distribution of CH_4 mass fraction for PHM and PRM.

Fig. 5 shows that CH_4 conversion for PHM and PRM at the reformer outlet is almost the same. However, for the cases of the particle-resolved model, zones with low concentrations of methane inside the catalyst particles can be observed because transport phenomena via convection are significantly faster than diffusion inside porous media. Therefore, local low methane concentration zones can be detected. One of the main reasons for this observation is that for industrial catalyst particles, the surface of the catalyst particle has a predominant effect compared to the mass of the catalyst. This was confirmed by Pedernera et al. [50], who reported that for large catalyst particles, steam methane reforming reactions occur in a zone close to the particle surface.

A quantitative comparison of numerical results for methane conversion at various residence times and temperatures, as well as a comparison with experiments, is shown in Fig. 6. Fig. 6 depicts that for all cases, the experimental CH_4 conversion is lower than that obtained via numerical simulation. At the same time, for all cases, CH_4 conversion obtained via PHM is higher than that obtained via PRM. An increase in residence time at 1000 K and 1200 K leads to a decrease in deviation between the numerical results obtained via PRM and experimental data, while at 800 K it leads to an increase in deviation. A detailed explanation of the possible reasons for such behavior is provided below in a separate subsection.

With heat supply through the wall

The contours of the methane mass fraction obtained via numerical simulations using the Particle-Resolved Model (PRM) and Pseudo-Homogeneous Model (PHM) are presented in Fig. 7. This figure shows that the methane mass fraction at the reformer outlet is significantly lower than that for the case without heat supply through the wall. The main reason for this is the endothermic nature of the steam methane reforming process. A heat flux through the wall compensates for the heat absorbed by the steam methane reforming reaction. Additionally, Fig. 7 demonstrates that CH_4 conversion for PHM is higher than that for PRM. A quantitative comparison of CH_4 conversion between numerical results and experimental data is provided below.

The comparison of numerical results for both cases, with and without heat supply, is depicted in Fig. 8. This figure clearly indicates that heat supply through the wall significantly increases methane conversion. The CH_4 conversion for the case without heat supply is 0.2319, while that for the case with heat supply is 0.4834.

The quantitative comparison of the numerical results obtained for PRM and PHM with experimental data is shown in Fig. 9. This figure illustrates CH_4 conversion versus residence time for various temperatures with heat supply through the wall: solid line – pseudo-homogeneous model; dashed line – particle-resolved model; hollow points – experimental data; dotted line – equilibrium methane conversion [51] for the given operational parameters (temperature, pressure, steam-to-methane ratio).

Fig. 9 reveals that PHM provides higher CH_4 conversion, while PRM yields lower CH_4 conversion than that obtained in the experiments. The deviation between numerical results obtained via PRM decreases as temperature and residence time increase. Moreover, Fig. 9 shows that for T = 1200 K and high residence time, CH_4 conversion approaches the equilibrium value; PHM provides CH_4 conversion that is 1.8 % less than the equilibrium value at 1200 K and 300 $kg_{cat}{\cdot}s/kmol_{CH_4}$.

Additionally, Fig. 9 depicts that CH_4 conversion increases nonlinearly as residence time increases. Such behavior in the dependence of CH_4 conversion on residence time has been observed by many authors [32,41,52].

The discrepancies between the obtained results for methane conversion via various methods can be explained by variations in temperature within the packed bed.

3.2. Temperature

Temperature is one of the key operational parameters that has a notable effect on methane conversion. The numerical simulation revealed a significant deviation in the nature of temperature contours for the cases with and without heat supply through the wall. Fig. 10 shows the temperature contours obtained via the Pseudo-Homogeneous Model (PHM) (a, c) and the Particle-Resolved Model (PRM) (b, d). The temperature contours for the case without heat supply through the wall are similar for both PHM and PRM because the heat of the reaction mixture is absorbed by the chemical reaction, and there is no external heat supply to the reaction mixture. The heat of the reaction mixture is transferred to catalytic particles mainly due to convection. Therefore, the outlet

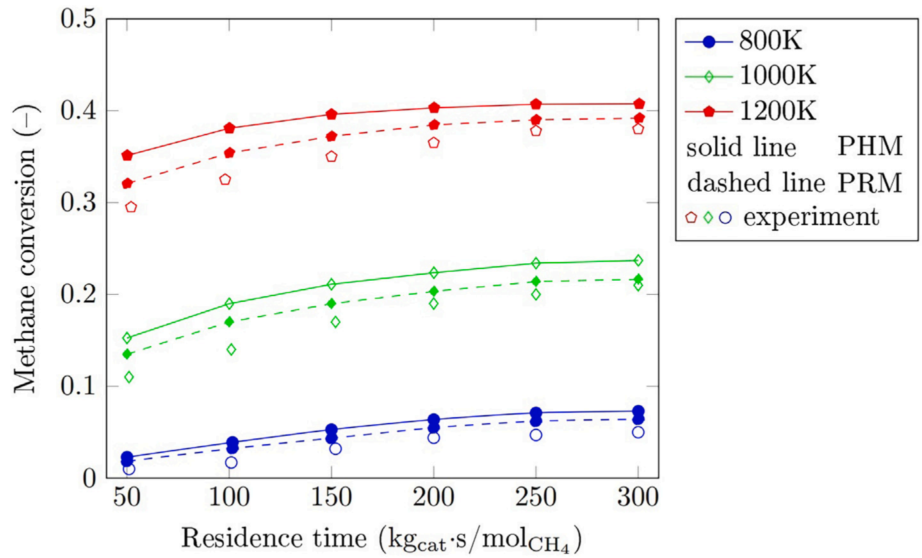


Fig. 6. The methane conversion vs. residence time for various temperatures without heat supply through the wall: solid line – pseudo homogeneous model; dashed line – particle resolved model; hollow points – experimental data.

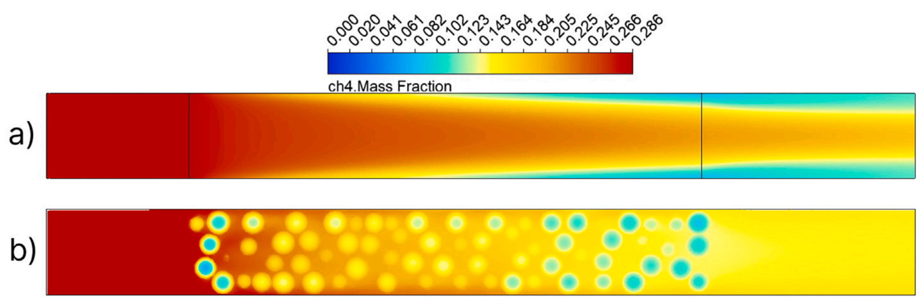


Fig. 7. The contours of CH_4 mass fraction obtained via PHM (a) and PRM (b) at T = 1000 K, $\beta = 2.0$, and a residence time of 200 $kg_{cat}\cdot s/kmol_{CH_4}$.

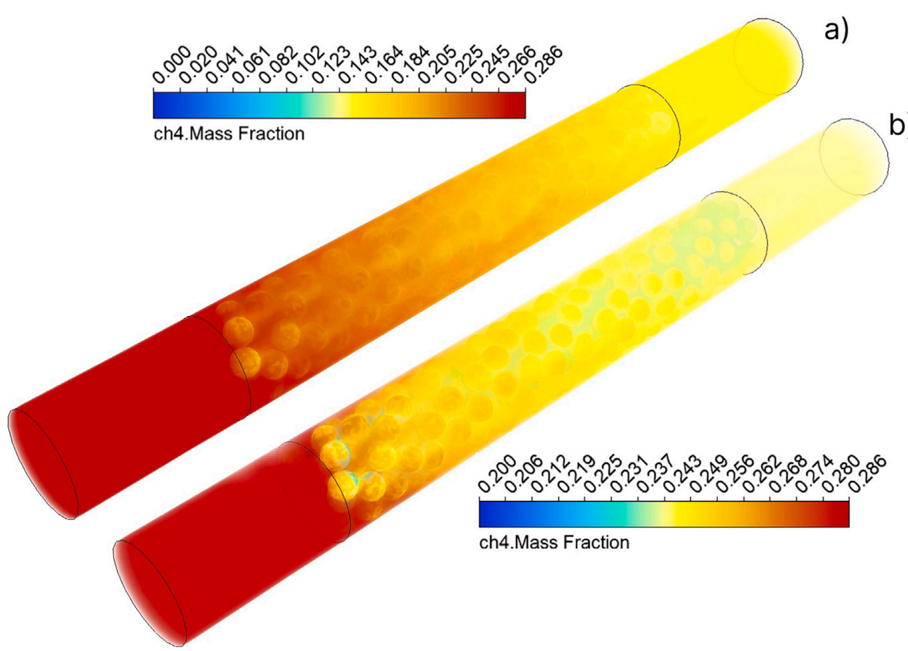


Fig. 8. The contours of CH_4 mass fraction obtained via PRM for the cases with heat supply (a) and no heat supply (b) at T = 1000 K, $\beta = 2.0$, and a residence time of 200 $kg_{cat}\cdot s/kmol_{CH_4}$.

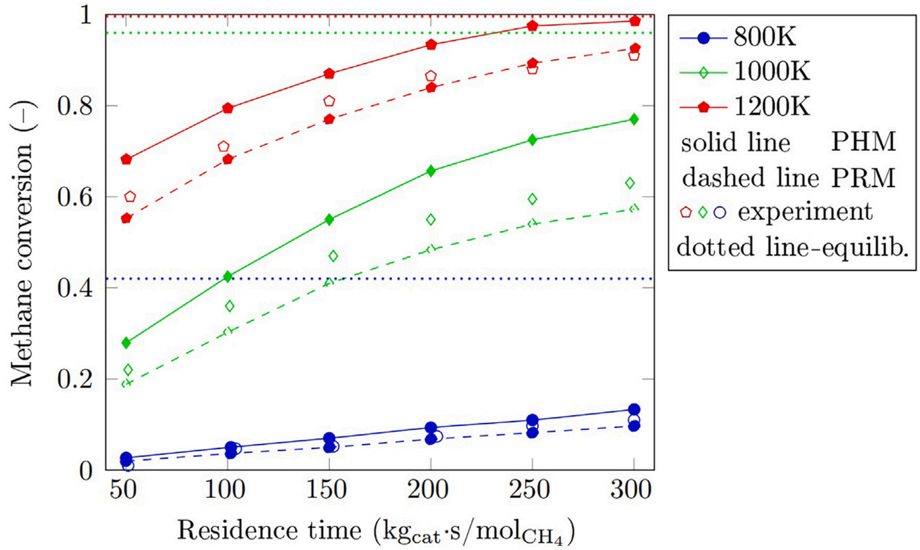


Fig. 9. The methane conversion vs. residence time for various temperatures with heat supply through the wall: solid line – pseudo homogeneous model; dashed line – particle resolved model; hollow points – experimental data; dotted line – equilibrium methane conversion for the given operational parameters (temperature, pressure, steam-to-methane ratio).

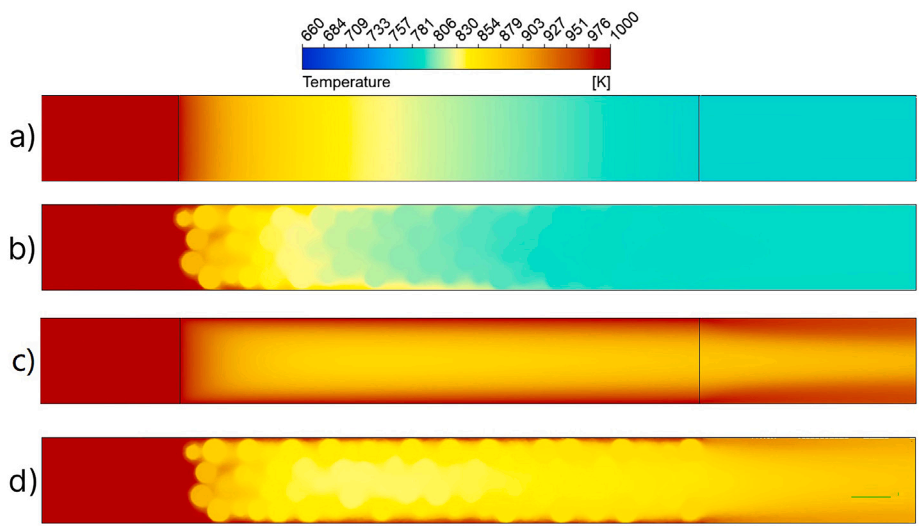


Fig. 10. The temperature contours for the cases without (a, b) and with (c, d) heat supply through the wall at T = 1000 K, $\beta = 2.0$, residence time of 200 $kg_{cat}\cdot s/kmol_{CH_4}$ obtained via PHM (a, c) and PRM (b, d).

temperatures for PHM and PRM are close to each other, at 773 K and 761 K, respectively.

However, the nature of temperature contours for the case with heat supply is completely different because the outlet temperatures for PHM and PRM are 899 K and 798 K, respectively. This significant deviation indicates a notable difference (about 20 %) in CH_4 conversion obtained for PHM and PRM, as shown in Fig. 6. One reason for this observation is the difference in the nature of heat transfer from the hot wall to the packed bed in PHM and PRM. In PHM, heat from the wall is transferred to the packed bed via thermal conductivity because it is treated as an isotropic porous region. In contrast, in the particle-resolved model, heat from the hot wall is transferred via convection and radiation (from the wall to the reaction mixture and then from the reaction mixture to the catalyst) as well as from the wall to particles via thermal conductivity at contact points. However, as mentioned above, to avoid errors in mesh generation, the catalyst particles were scaled down. Therefore, there are no contact points for PRM. As a result, the heat flux from the wall for PHM is higher than that for PRM. A quantitative comparison of outlet temperatures for various operational parameters is provided in Fig. 12.

Temperature contours for the steam methane reformer with (a) and without (b) heat supply through the wall at T = 1000 K, $\beta = 2.0$, and a residence time of 200 $kg_{cat}\cdot s/kmol_{CH_4}$ are presented in Fig. 11. Fig. 11 reveals that in the near-wall region, a local temperature increase occurs for the case with heat supply, and radial variation of temperature is significant. In contrast, for the case without heat supply, the radial deviation of temperature is negligible.

Fig. 12 shows the variation of outlet temperature with respect to residence time for the case without heat supply through the wall (a) and with heat supply through the wall (b).

Fig. 12a illustrates that an increase in residence time non-linearly decreases the outlet temperature. For example, for all investigated temperatures, the discrepancy between outlet temperatures for residence times of 200 and 250 $kg_{cat}\cdot s/kmol_{CH_4}$ is minimal. The main reason for this behavior is the effect of temperature on the reaction rate,

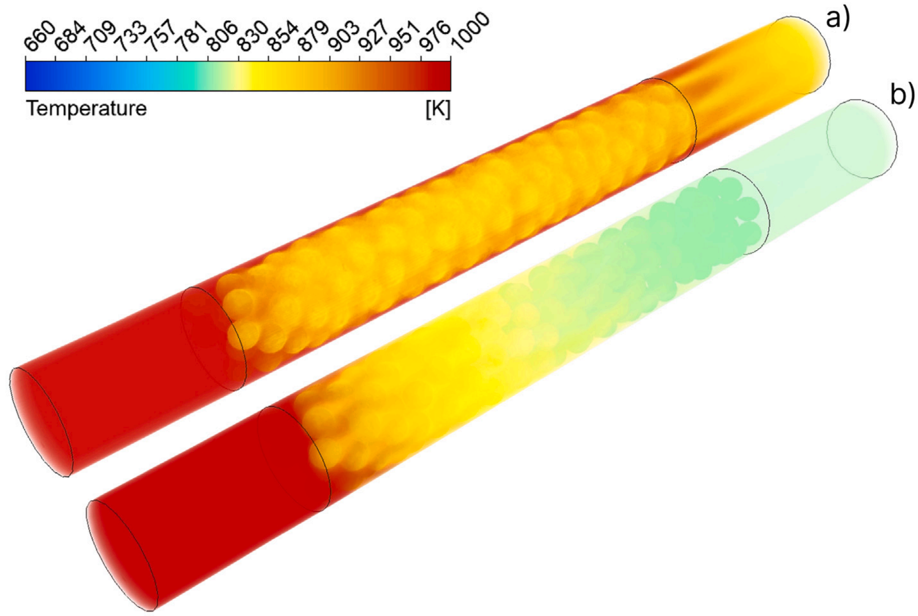


Fig. 11. Temperature contours for the steam methane reformer with (a) and without (b) heat supply through the wall at T = 1000 K, β = 2.0, residence time of 200 $kg_{cat}\cdot s/kmol_{CH_4}$.

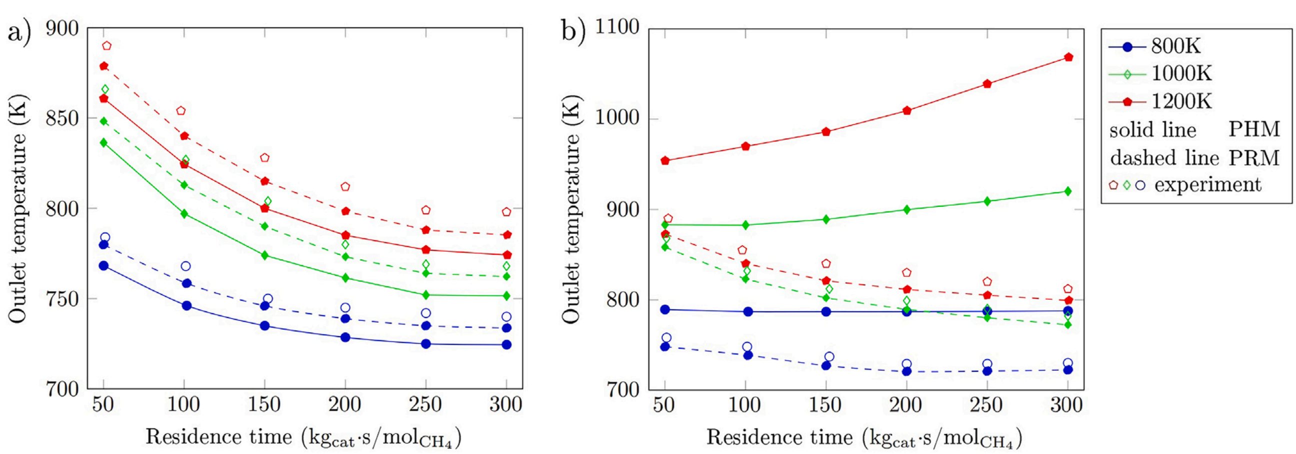


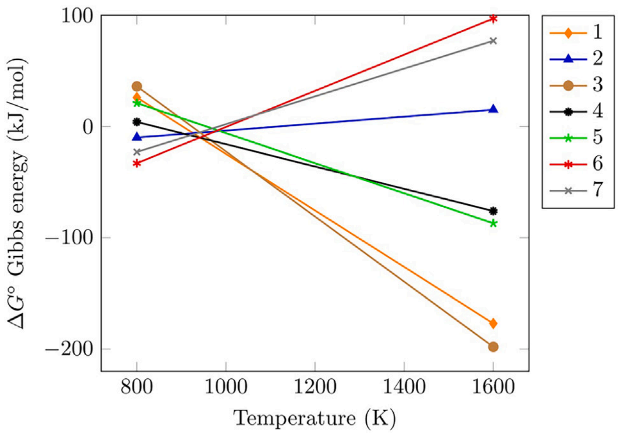
Fig. 12. Temperature at the reformer outlet vs. residence time for various inlet and wall temperatures: (a) without heat supply through the wall; (b) with heat supply through the wall. Solid line – pseudo homogeneous model; dashed line – particle resolved model; hollow points – experimental data).

which consequently affects the heat absorbed by endothermic reactions. The experimental results correlate well with those obtained via the Particle-Resolved Model (PRM). The outlet temperature obtained via the Pseudo-Homogeneous Model (PHM) is lower than that obtained from PRM, and one of the main reasons for this observation is the different nature of heat transfer in PHM compared to PRM. However, the discrepancy between the outlet temperatures for PHM and PRM is minimal because the limiting factors are the temperature of the reaction mixture and the enthalpy of the reaction mixture.

The dependence of outlet temperature on residence time has a completely different character for the case with heat supply through the wall, as shown in Fig. 12b. This figure indicates that an increase in residence time leads to a decrease in outlet temperature for PRM while resulting in an increase in outlet temperature for PHM. Moreover, an increase in the temperatures of both the inlet mixture and the wall leads to a greater temperature difference at high residence times. These observations can also be explained by the differing natures of heat transfer from the wall to packed bed as well as in the packed bed. The heat flux from the wall is higher for PHM than for PRM, and heat is transferred through the packed bed in PHM faster than in PRM. Therefore, at high residence times, after reaching close to equilibrium methane conversion, the heat flux is transferred to the packed bed to heat it up. The rapid increase observed at 1200 K can be attributed to the fact that CH4 has reached a nearly maximal conversion level ($X_{CH_4}\approx 1.0$), and the heat flux is used to increase the temperature. At the same, there is difference in the heat transfer inside packed bed. In PHM, the heat is transferred via thermal conductivity while in PRM the heat is transferred via convection and radiation.

The case for 800 K further clarifies this explanation. At this temperature level, methane conversion is minimal, and a notable portion of the heat supplied through the wall is used to heat up the reaction mixture, resulting in an outlet temperature close to 800 K for all residence times.

Fig. 12 shows that experimental results correlate well with numerical results obtained via PRM. The minimal difference can be explained by scaling down the computational domain, which resulted in changes to heat transfer at contact points. Therefore, the effective



No.	Reaction	ΔH_{298}, (kJ/mol)
	Methane oxidation	
1	$CH_4 + H_2O \Leftrightarrow CO + 3H_2$	206.4
2	$CO + H_2O \Leftrightarrow CO_2 + H_2$	−41.1
3	$CH_4 + CO_2 \Leftrightarrow 2H_2 + 2CO$	247.3
	Carbon formation	
4	$CH_4 \Leftrightarrow C + 2H_2$	74.8
5	$2CO \Leftrightarrow C + CO_2$	−173.3
6	$CO + H_2 \Leftrightarrow C + H_2O$	−90.13
7	$CO_2 + 2H_2 \Leftrightarrow C + 2H_2O$	−131.3

Fig. 13. The change in Gibbs free energy [59,60].

thermal conductivity of the actual packed bed is higher than that of the scaled-down packed bed used in this simulation.

3.3. Additional explanations of the results

The temperature has a significant effect on the reaction rate and, consequently, on methane conversion. The temperature distribution inside the packed bed for the case without heat supply through the wall is almost the same for the Pseudo-Homogeneous Model (PHM) and the Particle-Resolved Model (PRM) because there is no heat flux from the wall to the packed bed. The temperature inside the packed bed decreases non-linearly along the axial direction. The outlet temperature is determined by the total endothermic effect of the reactions, which, in turn, depends on methane conversion. A decrease in temperature within the packed bed leads to a decrease in CH_4 conversion, which subsequently reduces the amount of heat absorbed by endothermic reactions. Therefore, the temperature difference between the inlet and outlet increases when the inlet temperature increases.

Completely different results were observed for the case with heat supply through the wall. An increase in temperature and residence time leads to an increase in the difference between outlet temperatures obtained via PHM and PRM. One of the main reasons for this behavior is the different heat transfer mechanisms from the wall to the packed bed for PHM and PRM. For PRM, heat is transferred via convection and radiation from the wall to the packed bed, as well as through thermal conductivity at the contact points between the wall and catalyst particles. However, the mesh for PRM was scaled down, resulting in no contact points in our PRM model. For PHM, heat is transferred to the packed bed by thermal conductivity because the packed bed is treated as a porous medium. The results are plausible that the wall effects of the reactor are not captured by the PHM [35,53]. The porous media model incorporates an empirically determined flow resistance in the packed bed. Thus, this model is employed by adding a momentum sink in the governing momentum equations.

The effective thermal conductivity in the porous medium, k_{eff}, is computed by ANSYS FLUENT as the volume average of the fluid conductivity and the solid conductivity:

$$k_{\text{eff}} = \epsilon_p k_f + (1 - \epsilon_p) k_s \tag{20}$$

where k_f, k_s – fluid phase and solid medium thermal conductivity, respectively; ϵ_p – porosity of packed bed.

This expression shows a parallel model that predicts the theoretical maximum of effective thermal conductivity as a function of porosity. However, since the packed bed has a randomly packed structure, this parallel model may provide inaccurate predictions for the thermal conductivity of porous media [54,55]. There are many other models for predicting effective thermal conductivity, and assessing their applicability is crucial for the numerical simulation of steam methane reforming via the pseudo-homogeneous model.

Another factor that should be taken into account in the heat transfer mechanism is radiative heat transfer. In this paper, P1 radiation model is used. The P-1 radiation model is the simplest case of the more general P-N model, which is based on the expansion of the radiation intensity into an orthogonal series of spherical harmonics [56]. Therefore, an increase in temperature leads to an increase in the radiative heat transfer.

Carbon formation is another factor that needs to be considered in steam methane reforming. Experimental tests were performed using a lab-scale furnace with control over all parameters, such as steam-to-methane ratio, temperature, and pressure. The tests were conducted for a relatively short time on stream compared to industrial cases. Minor carbon traces on catalyst particles were observed only at 1200 K in the near-wall region close to the reformer inlet. Additionally, there is sufficient information in research papers indicating that carbon formation can be omitted for steam-to-methane ratios above 1.0 over coke-resistant catalysts [57,58].

Furthermore, we analyzed changes in Gibbs free energy for various possible chemical reactions in steam methane reforming, including carbon formation. Fig. 13 shows that in the temperature range above 1000 K, the chemical reaction of steam methane reforming is more favorable from a thermodynamic point of view than reactions leading to carbon formation, such as methane cracking and the Boudouard reaction.

Therefore, it can be concluded that in the present study, carbon formation and deposition have a negligible effect on material and energy balances. As a result, the chemical kinetic model considers only the reaction of steam methane reforming coupled with the water–gas shift reaction.

4. Conclusion

This paper presents a comparative analysis of CFD modeling of steam methane reforming within a packed bed containing spherical Ni-based catalyst particles, utilizing both the pseudo-homogeneous model (PHM) and the particle-resolved model (PRM). Experimental tests were conducted to evaluate the accuracy of the analyzed models. It was found that PHM requires significantly lower computational resources compared to PRM. A strong agreement was observed between the numerical results (CH_4 conversion and outlet temperature) obtained from both models and the experimental data for the case with an adiabatic reformer wall (without heat supply through the wall). For a residence time of 200–300 $\text{kg}_{\text{cat}}\cdot\text{s/kmol}_{CH_4}$, the deviation between the results was less than 10 %. The temperature contours for both PHM and PRM in the adiabatic wall case exhibited minimal deviation along the axial direction. Thus, PHM can be used for numerical simulation of SMR without heat supply through the wall to obtain accurate results.

Numerical results obtained via PRM for the case with heat supply showed good agreement with experimental data. The findings indicated that near-equilibrium methane conversion can be achieved for residence times greater than 200 $kg_{cat}\cdot s/kmol_{CH_4}$ at temperatures above 1200 K. However, CFD simulations for the case with heat supply through the wall demonstrated significant deviations in temperature and CH_4 mass fraction between PHM and PRM results. An increase in temperature and residence time corresponded with an increase in deviation between outlet temperatures, with maximum deviations exceeding 200 K for the analyzed parameters. The underlying cause of these deviations was identified as the difference in heat transfer mechanisms between PHM and PRM. In PHM, heat transfer occurs primarily through thermal conductivity, whereas in PRM, it involves convection, radiation, and thermal conductivity. PHM with parallel model for thermal conductivity does not accurately predict results for the case with heat supply through the wall. Consequently, it is concluded that PHM should be refined using an expression for effective thermal conductivity, as the standard parallel model does not accurately predict this parameter. Conversely, PRM effectively captures the performance of the steam methane reforming process. Further investigation into cases with constant heat flux through the wall is essential for a deeper understanding of the differences between PHM and PRM.

CRediT authorship contribution statement

Viacheslav Papkov: Writing – review & editing, Visualization, Validation, Software, Investigation, Formal analysis. **Dmitry Pashchenko:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data were used for the research described in the article.

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